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MATH 544

## Material

Background: 220/420/511

(1) Linear Systems

1.1 – 1.9 Direct methods

2.1 – 2.9 Sensitivity & perturbations

7.1 – 7.6 Iterative methods

(2) Least-squares | Singular Value Decomposition (SVD)

3.1 – 3.6 (in detail)

4.1 – 4.4

(3) Eigenvalues & Eigenvectors

5.1 – 5.9 (selectively)

6.1 – 6.7

(4) Nonnegative matrix theory (entrywise)

Applications: Google algorithm

Nonn. Matrix Factorization

# Notation & conventions

- $\mathbb{R}^n$, $\mathbb{C}^n$ : n-vectors (as columns) from $\mathbb{R}$ (reals) or $\mathbb{C}$ (complex)

- $M_{m,n}(\mathbb{R})$ : $m \times n$ matrices over $\mathbb{R}$

  $M_n(\mathbb{R})$ : $n \times n$ — " —
  similarly for $\mathbb{C}$.

- $A \in M_{m,n}(\mathbb{R}) \rightsquigarrow A = [a_1 | a_2 | \dots | a_n]$

  $a_j$ = j-th column in $\mathbb{R}^m$

  $$= \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_m^T \end{bmatrix}$$

  $v_k$ : k-row of $A$ in $\mathbb{R}^n$

  ( $A^T$ : transpose of $A$ )

- <u>Submatrices</u> $A \in M_n(\mathbb{R})$, $A = [a_{ij}]$

  Given a set $\alpha \subseteq \{1, 2, \dots, n\}$ $(\neq \emptyset)$
  ~~Then~~ we consider the elements of $\alpha$ in increasing order.

  By $A[\alpha]$ we denote the (<u>principal</u>) <u>submatrix</u> of $A$ obtained from the intersection of rows $i$ & columns $j$, $i, j \in \alpha$

E.g.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\alpha_1 = \{1, 3\}$$

$$\alpha_2 = \{2, 3\}$$

$$\alpha_3 = \{1\}$$

$$\alpha_4 = \{1, 2, 3\}$$

$$\cancel{\alpha_5 = \{\}}$$

$$A[\alpha_1] = \begin{bmatrix} 1 & 3 \\ 7 & 9 \end{bmatrix}$$

$$A[\alpha_2] = \begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix}$$

$$A[\alpha_3] = \begin{bmatrix} 1 \end{bmatrix}$$

$$A[\alpha_4] = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

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Quick discussion of first few sections in Chapter 1.

## § 1.1 Matrix multiplication

Matrices are real & square $A \in M_n(\mathbb{R})$

$\mathbb{R}$ is stored in ~~floating~~ floating point format (... FLOPS) – details in Chapter 2.

Say $A \in M_n(\mathbb{R})$, $x \in \mathbb{R}^n$ and want to find $Ax$:

$$Ax = b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$A = [a_{ij}]$$

$$b_i = \sum_{j=1}^{n} a_{ij} x_j = a_{i1} x_1 + a_{i2} x_2 + \dots$$

So the process of computing $b_i$ $(i = 1, 2, \dots, n)$

$$b_i \longleftarrow b_i + a_{ij} x_j, \quad i = 1, 2, \dots, n; \quad j = 1, 2, \dots, n$$

replacement (update)

addition & multiplication 2 FLOPS

TOTAL FLOPS FOR $Ax$:

$$\sum_{i=1}^{n} \sum_{j=1}^{n} 2 = 2n^2$$

How about multiplying two matrices in $M_n(\mathbb{R})$

$$\underset{n\times n}{A}\ \underset{n\times n}{B} = A\,[\,b_1\ b_2 \dots b_n\,]$$

columns of B

$$= [\,Ab_1\ Ab_2 \dots Ab_n\,]$$

which needs $n(2n^2) = 2n^3$ FLOPS

We say matrix $\times$ vector is $O(n^2)$ operations
matrix $\times$ matrix is $O(n^3)$ — " —

<u>Improvement</u> (reduces fetching of data & exploits multiple processors)

Consider submatrices of A, B

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

Partition A, B in blocks:

"conformal" partition of A, B
$$\left\{ \begin{array}{l} A_{11} = A[\{1,2,\dots,k\}] \\ B_{11} = B[\{1,2,\dots,k\}] \end{array} \right.$$

"leading" principal submatrices

$B_{12}$, $A_{12}$ are $k\times(n-k)$
$B_{21}$ $A_{21}$ are $(n-k)\times k$
$A_{22}$, $B_{22}$ are $(n-k)\times(n-k)$

Then $AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$

$$= \begin{bmatrix} \overset{k \times k}{A_{11} B_{11} + A_{12} B_{21}} & A_{11} B_{12} + A_{12} B_{22} \\ A_{21} B_{11} + A_{22} B_{21} & A_{21} A_{22} + B_{12} B_{22} \end{bmatrix}$$

"block multiplication"

Idea: Send each $\boxed{\text{block}}$ to a separate processor.

Speed up multiplication, but FLOPS are the same! Code looks different.

("Google" "fast matrix multiplication" to see state of the art acceleration)

## §1.2 Systems of Linear Equations

Solve $\underset{n\times n}{A}\ \underset{n\times 1}{x} = \underset{n\times 1}{b}$

$A \to$ coeff. matrix

$b \to$ right-hand side

$x \to$ unknown

Given $A, b$, "find" $x$

Focus: on square systems with unique solution

This means

$A$ is invertible (aka nonsingular)

$A^{-1}$ exists

$AA^{-1} = A^{-1}A = I = I_n \to n\times n$ identity.

**Theorem** Let $A \in M_n(\mathbb{R})$. TFAE

(a) $A^{-1}$ exists

(b) There is no $y \neq 0$ such that $Ay = 0$

(c) Columns of $A$ are linearly independent

(d) Rows of $A$ are linearly independent

(f) $\forall b \in \mathbb{R}^n$, $Ax = b$ has unique solution $x = A^{-1}b$

(g) $\det A \neq 0$

(h) No eigenvalue of $A$ equals $0$

## §1.3 Triangular systems

$G = [g_{ij}] \in M_n(\mathbb{R})$ is called <u>lower triangular</u> if $g_{ij} = 0$ for all $j > i$

$$G = \begin{bmatrix} \ddots & & 0 \\ * & \ddots & \\ & & \ddots \end{bmatrix}$$

anything.

eg $\begin{bmatrix} 1 & 0 & 0 \\ 3 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

lower triangular part.

<u>Fact</u> If $G = [g_{ij}]$ is lower triangular

then $\det G = \prod_{i=1}^{n} g_{ii}$

Solve system when $G$ is <u>upper triangular</u>

$\boxed{Gy = b}$ G is lower tri., invertible



$$g_{11} y_1 = b_1 \Rightarrow y_1 = \frac{b_1}{g_{11}}$$

$$g_{21} y_1 + g_{22} y_2 = b_2 \Rightarrow y_2 = \frac{b_2 - g_{21} y_1}{g_{22}}$$

$$\vdots$$

$$y_i = \frac{b_i - \sum_{k=1}^{i-1} g_{ik} y_k}{g_{ii}}$$

$i = 1, 2, \ldots, n$

"Forward substitution" to solve the system.

## §1.3 Triangular systems

$G = [g_{ij}] \in M_n(\mathbb{R})$ is called <u>lower triangular</u> if $g_{ij} = 0$ for all $j > i$:

$$G = \begin{array}{c} \\ i \\ \\ \end{array} \begin{array}{c} j \\ \left[\begin{array}{ccc} & & 0 \\ & * & \\ & & \end{array}\right] \end{array} \quad \text{e.g. } \begin{bmatrix} 1 & 0 & 0 \\ 3 & -1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

(similarly: upper triangular)

<u>Fact</u> If $G = [g_{ij}]$ is lower triangular, then $\det G = \prod_{i=1}^{n} g_{ii}$

Solve a system $\boxed{Gy = b}$, $G \in M_n(\mathbb{R})$ is lower triangular:

$(g_{11} \neq 0)$ $\quad g_{11} y_1 = b_1 \Rightarrow y_1 = b_1 / g_{11}$

$(g_{22} \neq 0)$ $\quad g_{21} y_1 + g_{22} y_2 = b_2 \Rightarrow y_2 = \dfrac{b_2 - g_{21} y_1}{g_{22}}$

$\vdots$

$$\Rightarrow y_i = \frac{b_i - \sum_{k=1}^{i-1} g_{ik} y_k}{g_{ii}}$$

$i = 1, 2, \ldots, n$

Solved!

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(§1.3 continued)

This last method is known as

<u>forward substitution</u> :

```
for i = 1:n
    for j = 1:i-1
    if g_ii = 0, set ERROR | STOP
    b_i ← b_i - g_ij b_j
    b_i ← b_i / g_ii
    end
end
```

FLOPS : $\underset{\substack{\uparrow \\ \text{add/division}}}{2} \times (0+1+2+\cdots+(n-1))$

$$= 2\sum_{i=1}^{n}(i-1)$$

$$= 2\,\frac{n(n-1)}{2} = n(n-1)$$

$$= n^2 - n$$

So $O(n^2)$ algorithm

( n comparisons as well)

What if a system is <u>not</u> triangular?

(Will return to 1.4)

## § 1.7 Gaussian Elimination / LU-factorization

Solve $Ax = b$, $A \in M_n(\mathbb{R})$
by transforming this system into an "equivalent system" (has same solution set) and it's of the form

$$Ux = y, \quad U \in M_n(\mathbb{R})$$

$U$ is upper triangular

Three types of row operations on $A$ that produce equivalent systems:

(1) Add multiples of one row of $A$ to the other rows of $A$.

(2) Interchange rows of $A$.

(3) Multiply a row by a nonzero number (adjust right-hand side accordingly)

Example $\longrightarrow$

## Example

Solve
$$\begin{cases} 2x_1 + x_2 + x_3 = 1 \\ 2x_1 + 2x_2 - x_3 = 3 \\ 4x_1 - x_2 + 6x_3 = 4 \end{cases}$$

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 2 & 2 & -1 & 3 \\ 4 & -1 & 6 & 4 \end{array}\right], \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad Ax = b$$

$$\underbrace{\phantom{xxxxxx}}_{A} \quad \underbrace{\phantom{xx}}_{b}$$

$$\xrightarrow[R_3 - 2R_1]{R_2 - R_1} \left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 0 & 1 & -2 & 2 \\ 0 & -3 & 4 & 2 \end{array}\right] \xrightarrow{R_3 + 3R_2} \left[\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & -2 & 5 \end{array}\right]$$

$$\underbrace{\phantom{xxxxxx}}_{U} \quad \underbrace{\phantom{xx}}_{y}$$

Solve $\boxed{Ux = y}$

Backward substitution:

$$x_3 = \boxed{5/-2}$$

$$x_2 - 2x_3 = 2 \Rightarrow x_2 = 2 + 2x_3 = \boxed{-3}$$

$$2x_1 = 1 - x_2 - x_3 \Rightarrow \boxed{x_1 = \frac{13}{4}}$$

Row ops performed can be expressed by left matrix multiplications:

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{L_1} \underbrace{\begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & -1 \\ 4 & -1 & 6 \end{bmatrix}}_{A} = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & -2 \\ 4 & -1 & 6 \end{bmatrix} \quad (R_2 - R_1)$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}}_{L_2} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & -2 \\ 4 & -1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & -3 & 4 \end{bmatrix} \quad (R_3 - 2R_1)$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}}_{L_3} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & -3 & 4 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & -2 \end{bmatrix}}_{U} \quad (R_3 + 3R_2)$$

<u>So</u>

$$L_3 L_2 L_1 A = U$$

$$\Rightarrow \quad A = \underbrace{L_1^{-1} L_2^{-1} L_3^{-1}}_{L} U$$

<u>Facts</u> (prove)

1) inverse of inv. lower triangular is lower triangular

2) Product of lower triangular matrices is lower triangular

Thus $L = L_1^{-1} L_2^{-1} L_3^{-1}$ is triangular triangular

So we factored:

$$(*) \quad \boxed{A = \underbrace{L}_{\text{lower triang.}} \cdot \underbrace{U}_{\text{upper triangular}}}$$

To solve $\boxed{Ax = b}$:

$$\Rightarrow \quad LUx = b$$

$$\Rightarrow \quad \boxed{Ux = L^{-1}b}$$

The new system is equivalent to original & can be solved with backward substitution

We call $(*)$ the <u>LU-factorization of A</u>.

What allowed to achieve $(*)$:

(1) $Ax = b$ had unique solution ($\det A \neq 0$)

(2) We did not (have to) use row interchanges.
(would have to use ~~pe~~ interchanges if a pivot were $= 0$)

What is a condition (theoretically) for no need for permutations of rows:

If you look at our example:

$$\det [a_{11}] = a_{11} \neq 0$$

$$\det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \neq 0 \quad \text{(check)}$$

$$\det \underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}}_{A} \neq 0$$

<u>Fact</u> A admits LU factorization without row interchanges ("without pivoting") iff all leading principal submatrices of A are invertible.

Nevertheless ....